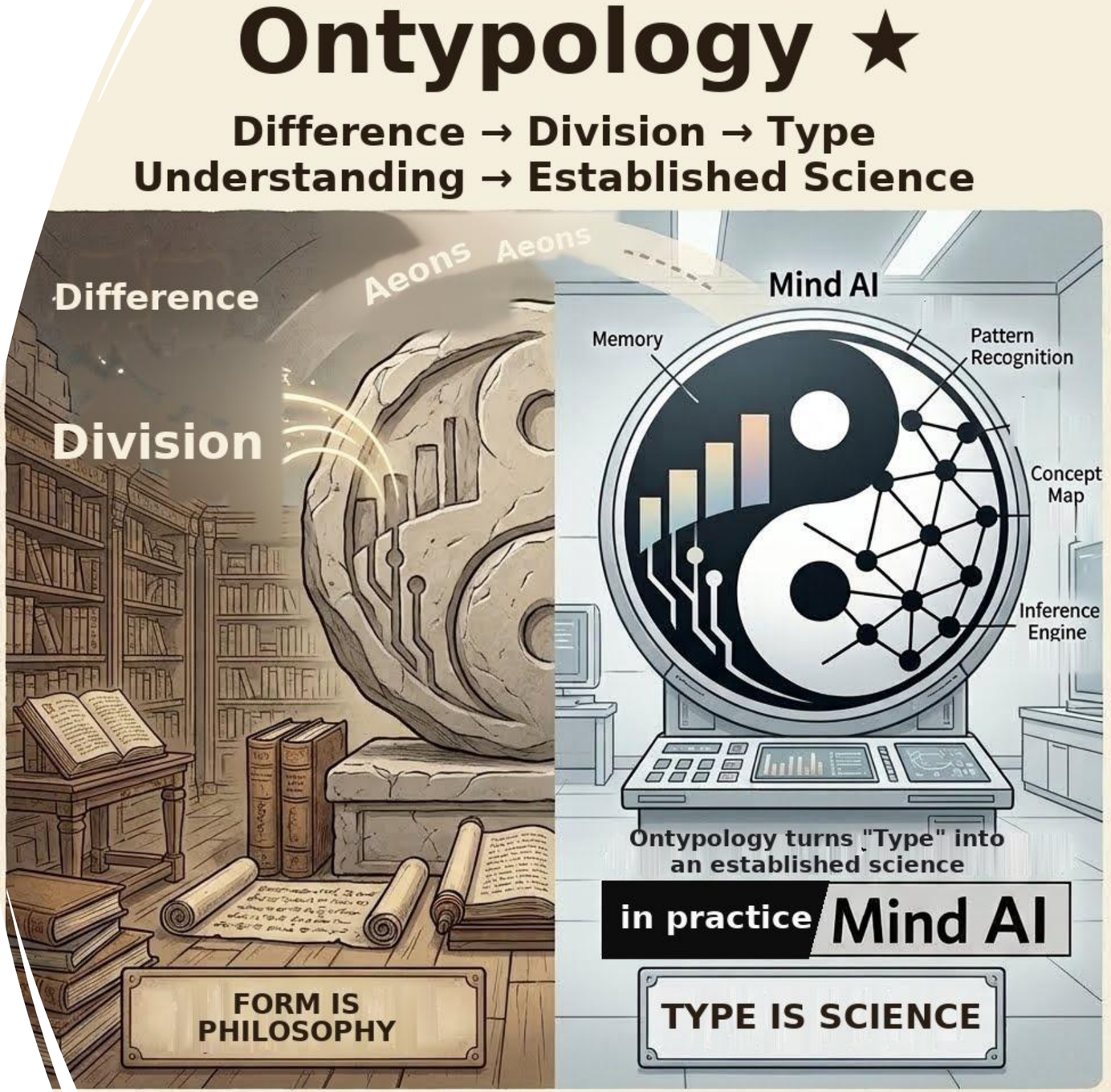
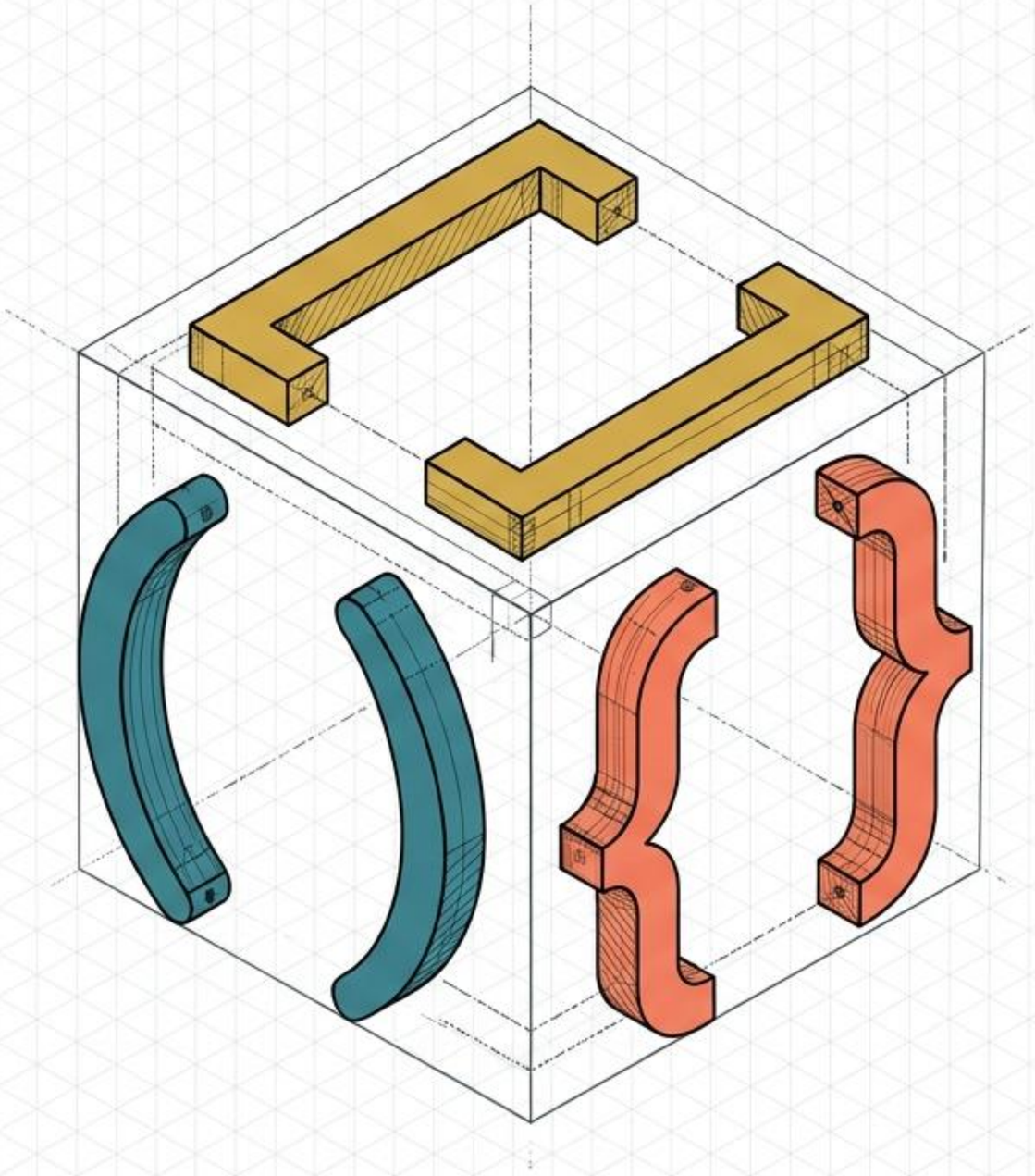
Ontypology

- From sameness and difference to separation and merge, from separation and merge to Type, from Type to understanding—the brain spent eons walking this path.
- A century ago, logic was philosophy; once seen clearly, it became a science.
- Today, exploring “Type” (a concept of the brain) is philosophy. Ontypology sees “Type” clearly, turns it into a science—and practices it as Mind AI.





The Precision Supplement for Mathematical Proof

Three Language Enhancements for Set Theory

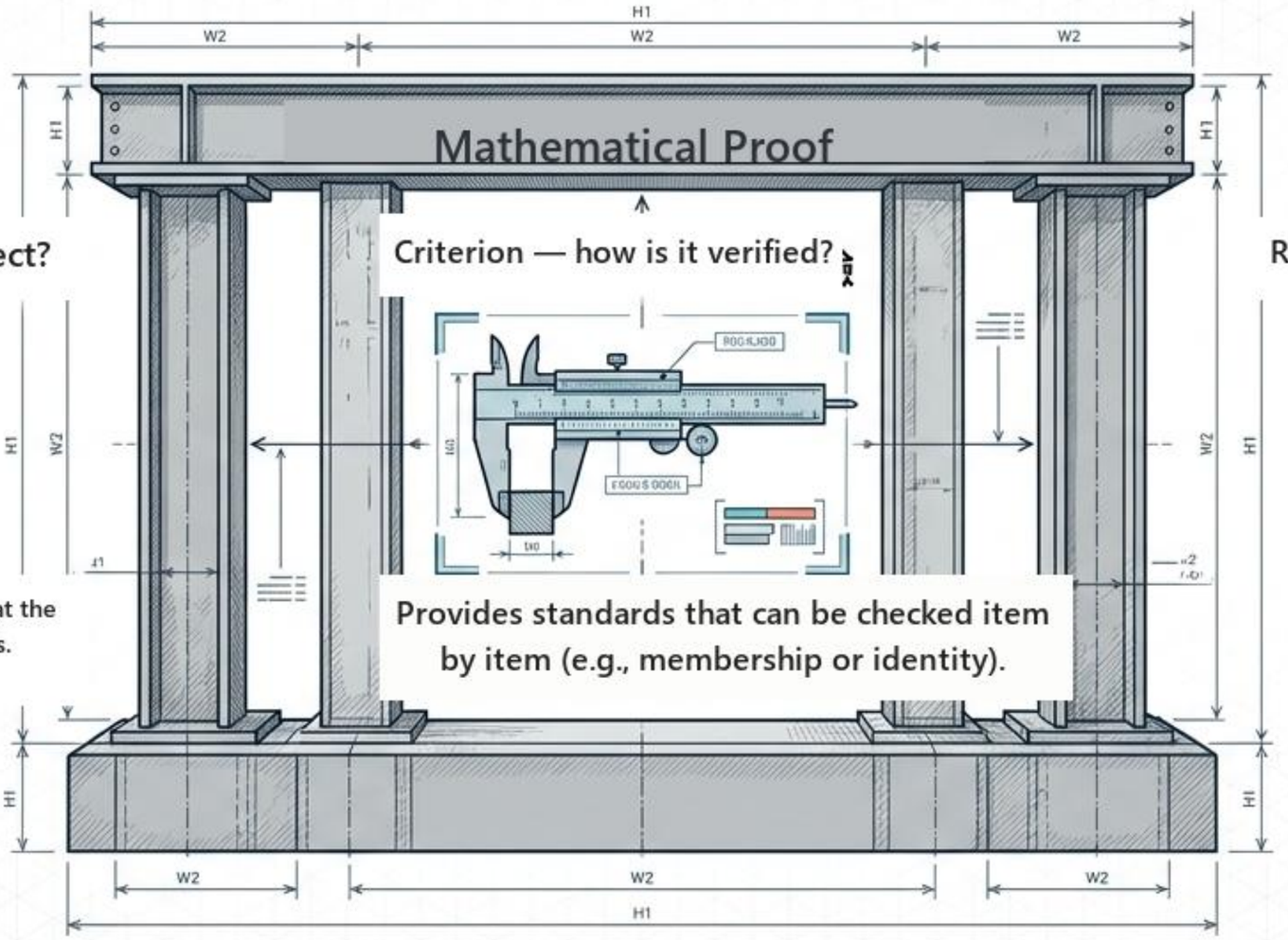
A complete illustrated analysis of “Type”.

Type = { } + ( ) + [ ]

```
[GLOSSARY DEFINITION]
Type (never Form).
A fully described mathematical object.
```

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The Three-Piece Set of Proof: The Foundation of Mathematical Logic



Definition — what is the object?



Draws a clear boundary: it defines what the category includes and what it excludes.

Criterion — how is it verified?

Provides standards that can be checked item by item (e.g., membership or identity).

Reasoning — how does it proceed?



Starts from the known and advances step by step to the conclusion, ensuring every step can be rechecked.

Pure set language pushes “in or out” to its limit — but for a proof to stand, we must restore the dimensions it leaves out.

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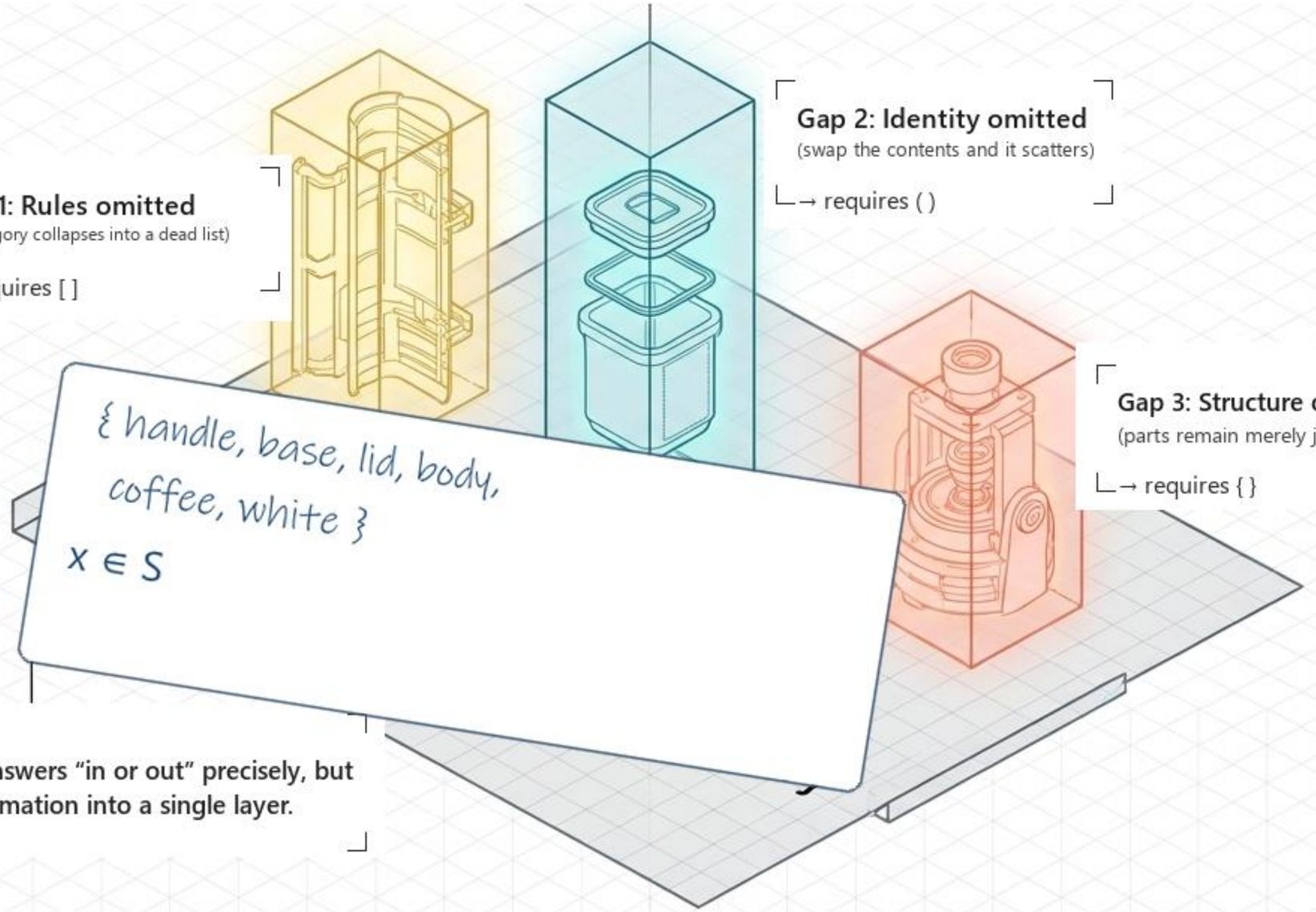
The Limit of Pure Set Language: Three Flattened Gaps

A proof's completeness comes from re-lifting the flattened information into three dimensions.

Gap 1: Rules omitted
(a category collapses into a dead list)
→ requires [ ]

Gap 2: Identity omitted
(swap the contents and it scatters)
→ requires ( )

Gap 3: Structure omitted
(parts remain merely juxtaposed)
→ requires { }



Pure set language answers "in or out" precisely, but presses all information into a single layer.

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The Supplement Map: Three Gaps, Three Dedicated Tools

The Gap (the gap)	Pure-Set Notation (flat)	Supplemented Notation (dimensional)	Dedicated Tool (the toolkit)
Rule	{ mug, glass } (closed list)	[x | x passes ∩(T)] (open rule)	[ ] Abstract Set
Identity	{ water } → { coffee } (set replaced)	(water) → (coffee) (id(C) unchanged)	( ) Container Set
Structure	{ handle, base } (mere juxtaposition)	remove(p, M) (three-level grading)	{ } Composition Set

The two languages do not replace each other — they stack: pure sets are the skeleton of a proof, and the three supplements are its flesh and blood.

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Tool 1: [ ] Abstract Set — Write a Category as a Rule

$$T = (F(T), \cap(T))$$

T: Abstract Category.

Example: [cup].

F(T): Feature Set.

Lists every feature the category requires. E.g. { can hold liquid, can be held and drunk from }.

∩(T): Intersection Criterion.

Stipulates that every feature must pass before the object belongs to the category.

The dead list

{ mug, glass, paper cup }

Cannot classify a newly invented thermos.

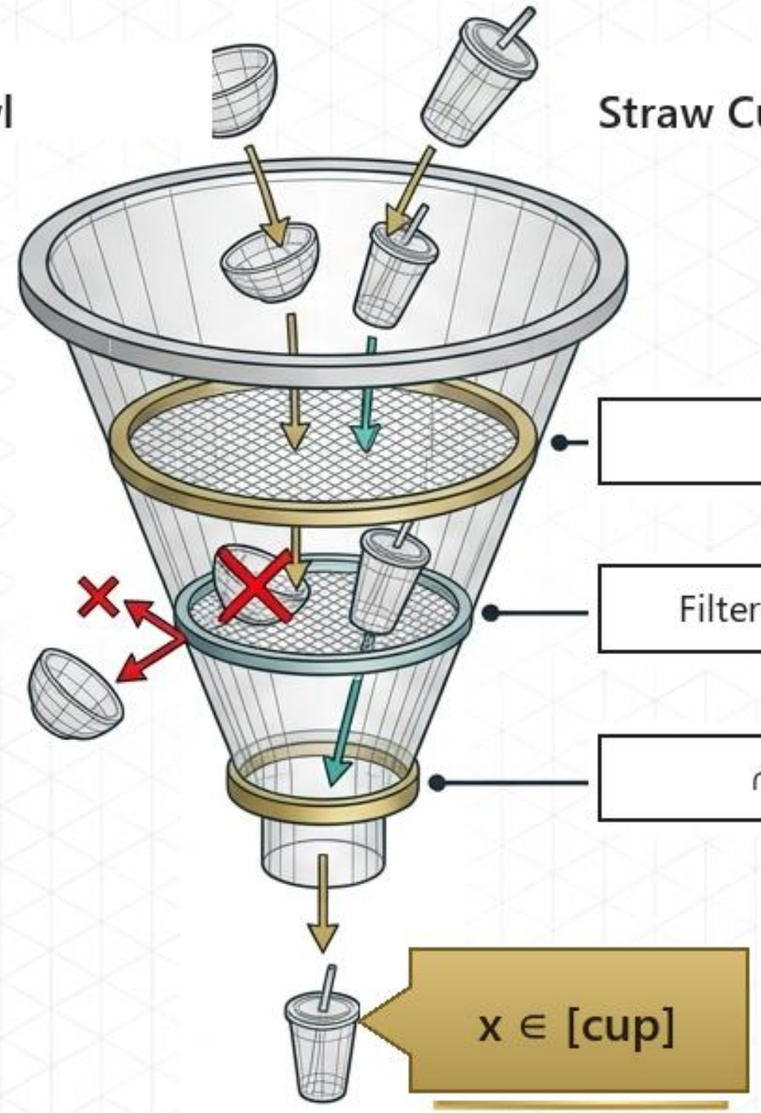
The open rule

[x | x passes ∩(T)]

Dynamically evaluates any new object based on features.

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The Intersection-Criterion Funnel: From Features to Automatic Classification



Process Steps

1. Determine F(T)
2. Test each f(x)
3. Take the intersection ∩(T)(x)
4. Classify automatically if all pass

Pro Tip: Borderline cases (e.g., a syringe) sharpen the definition; the choice of F(T) is a stance and must be written at the start of the proof.

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Tool 2: ( ) Container Set — Write Invariance as an Axiom

$$C = (id(C), S(C))$$

id(C): Identity marker.

An absolute tag that does not change when the contents change.



Anatomy of a Container

S(C): Inner space.

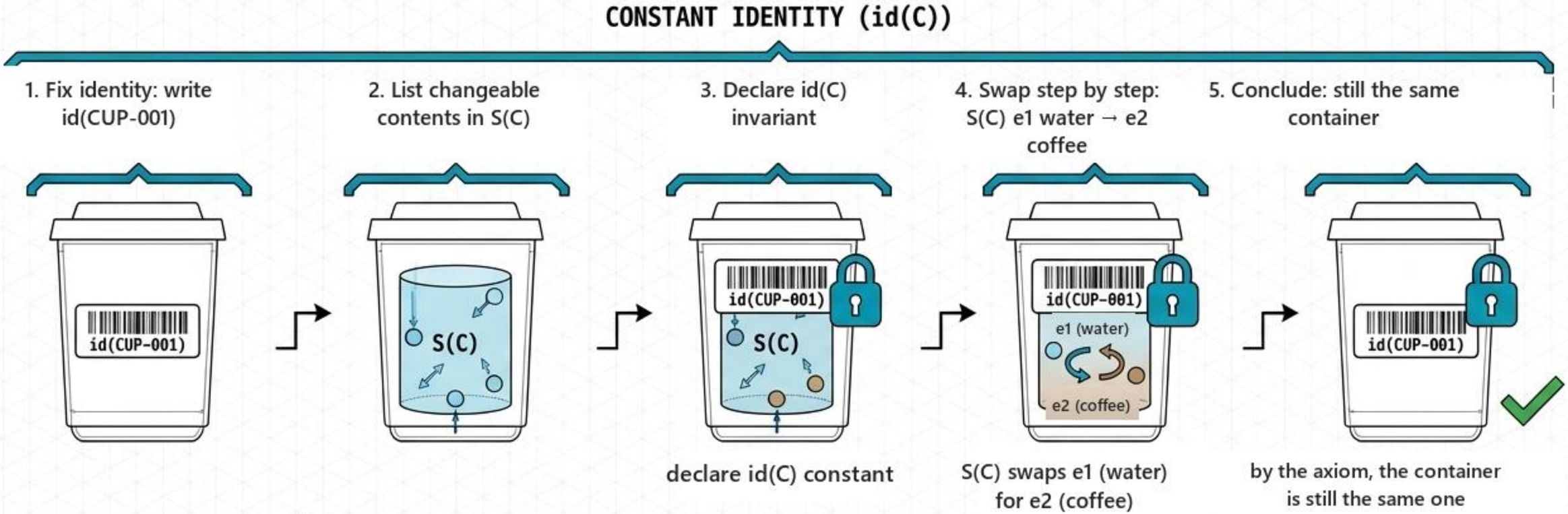
A set of elements that can be held, replaced, and emptied.

[Invariance Axiom] For any e1, e2 ∈ S(C), replacing e1 with e2 does not change id(C).

Establish identity first, then discuss contents!

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The Invariant State Machine: Proving Sameness Through Change



Diagnostic Contrast

FATAL ERROR: Identity Lost

{ water } → { coffee }

{ water } → { coffee } = Identity lost

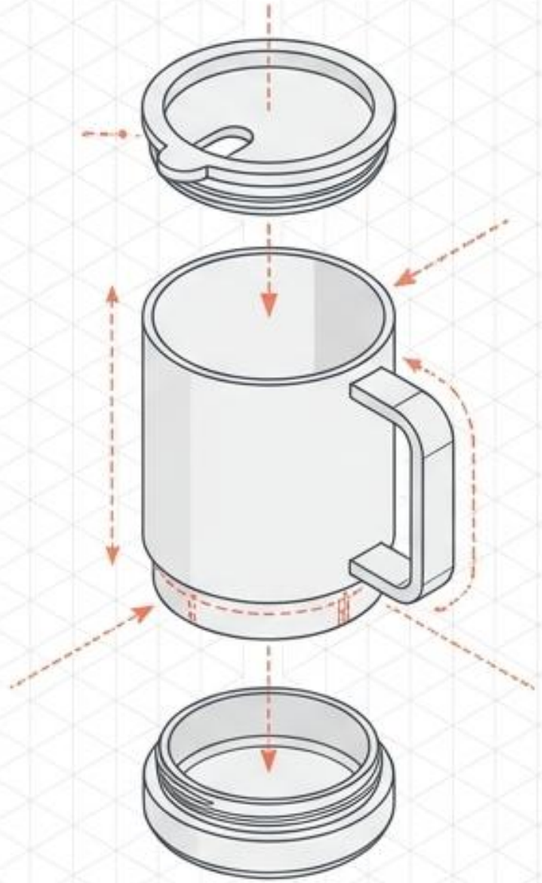
SUCCESS: Identity Maintained

(water) → (coffee)

(water) → (coffee) = Identity maintained

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Tool 3: { } Composition Set — Write Dependence as a Criterion



Core Question

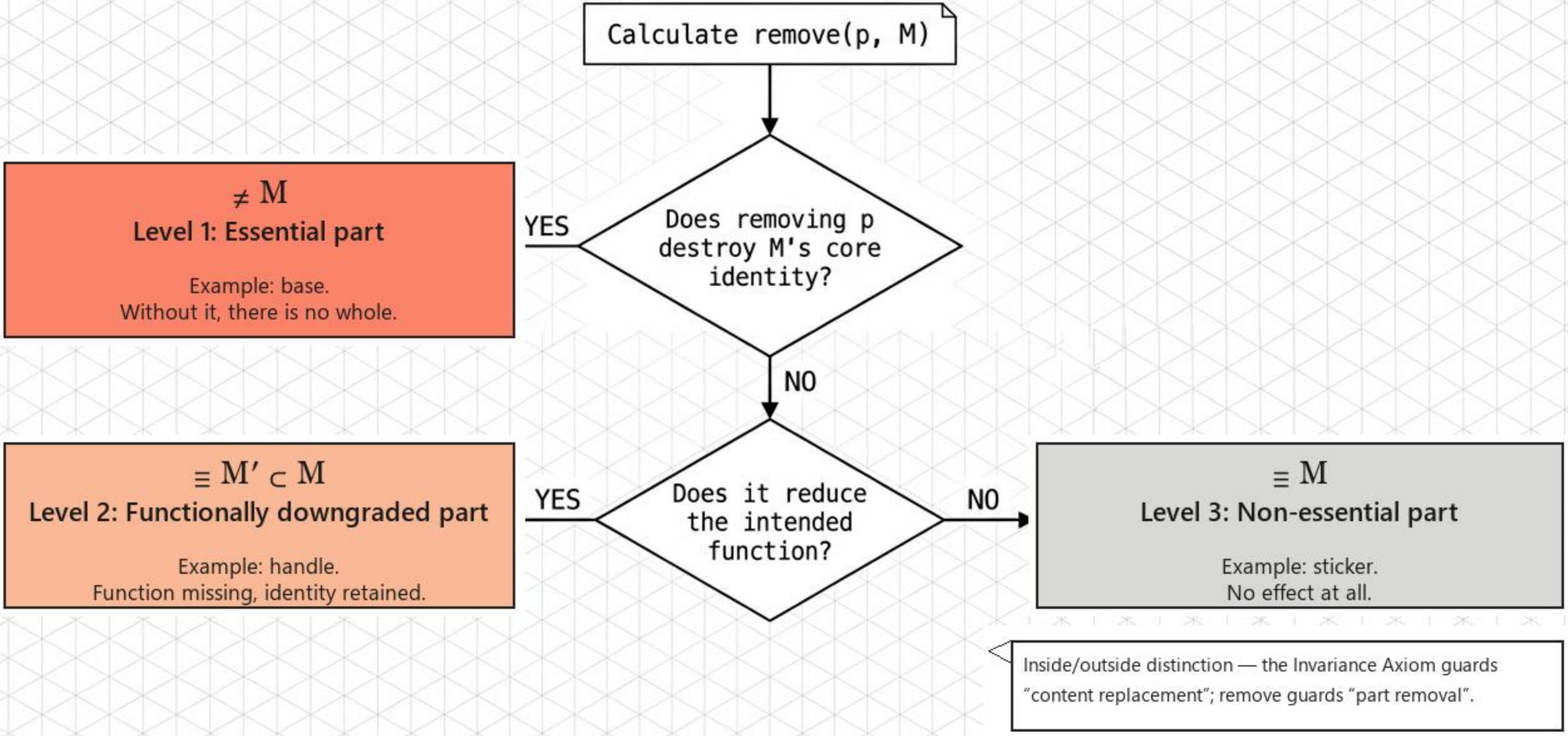
BOM criterion: “After taking it apart, does it still recognize itself?”

Passed (enters BOM)	Failed (enters property)
Lid: taken off, it is still a lid.	White (colour): taken off, it is nothing.
-> Structure { }	-> Property

A proof needs not a flat list of parts but a relation of dependence. This boundary separates “Structure” from “Appearance” completely.

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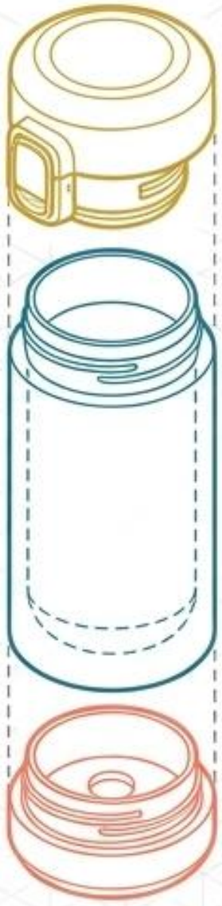
The remove-Criterion Decision Tree: Precisely Grading Dependence



Ultimate Integration: The Mathematical Equation of a Unit Type

Unit Type =

{ parts } + (contents) + [category]



[Category]
- The abstract features and rules it fulfills.

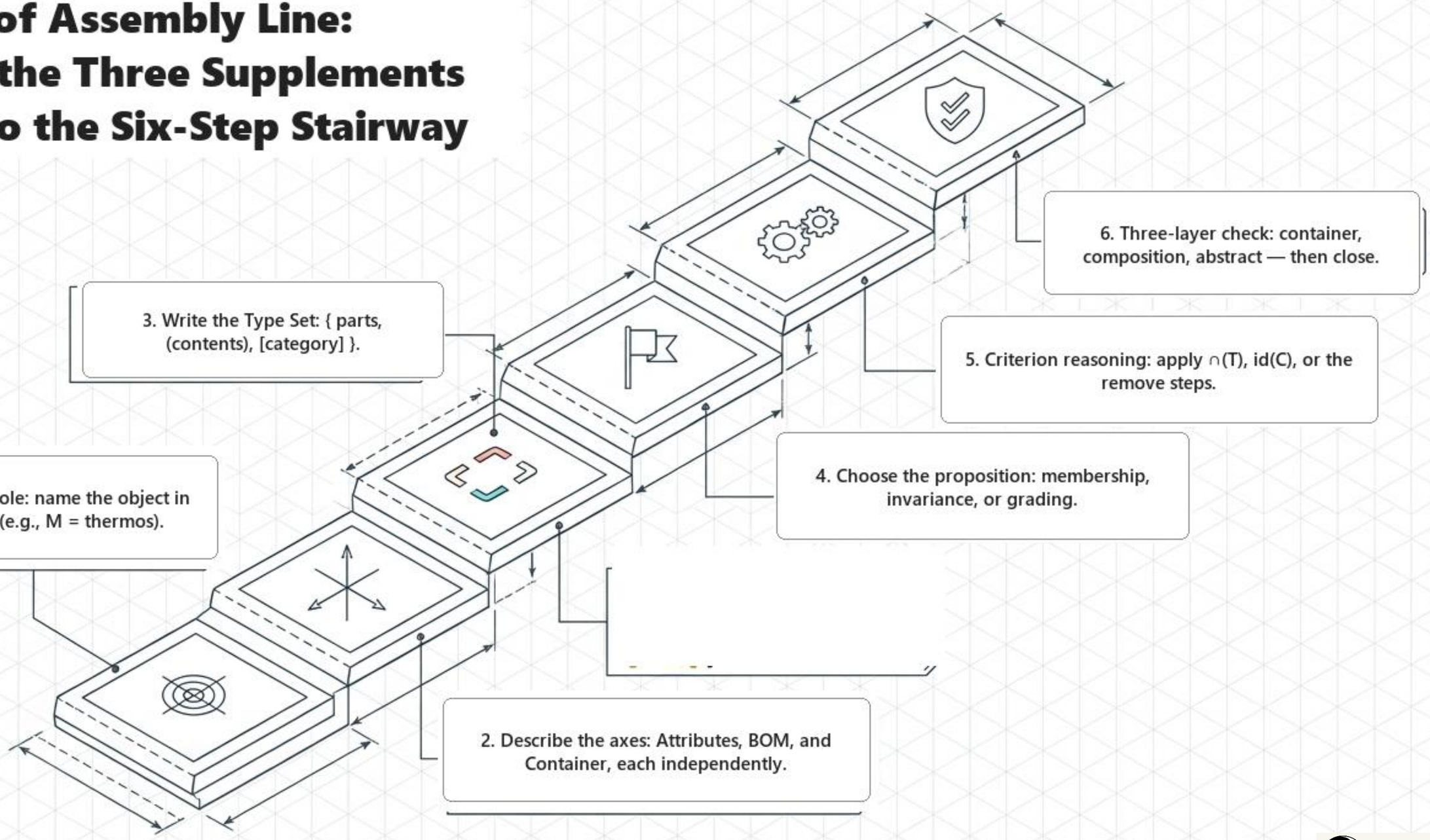
(Content)
- The internal state and constant identity.

{ BOM }
- The physical components (Structure).

Understanding a Type means precisely grasping its appearance, contents, composition, and dependence — none may be missing.

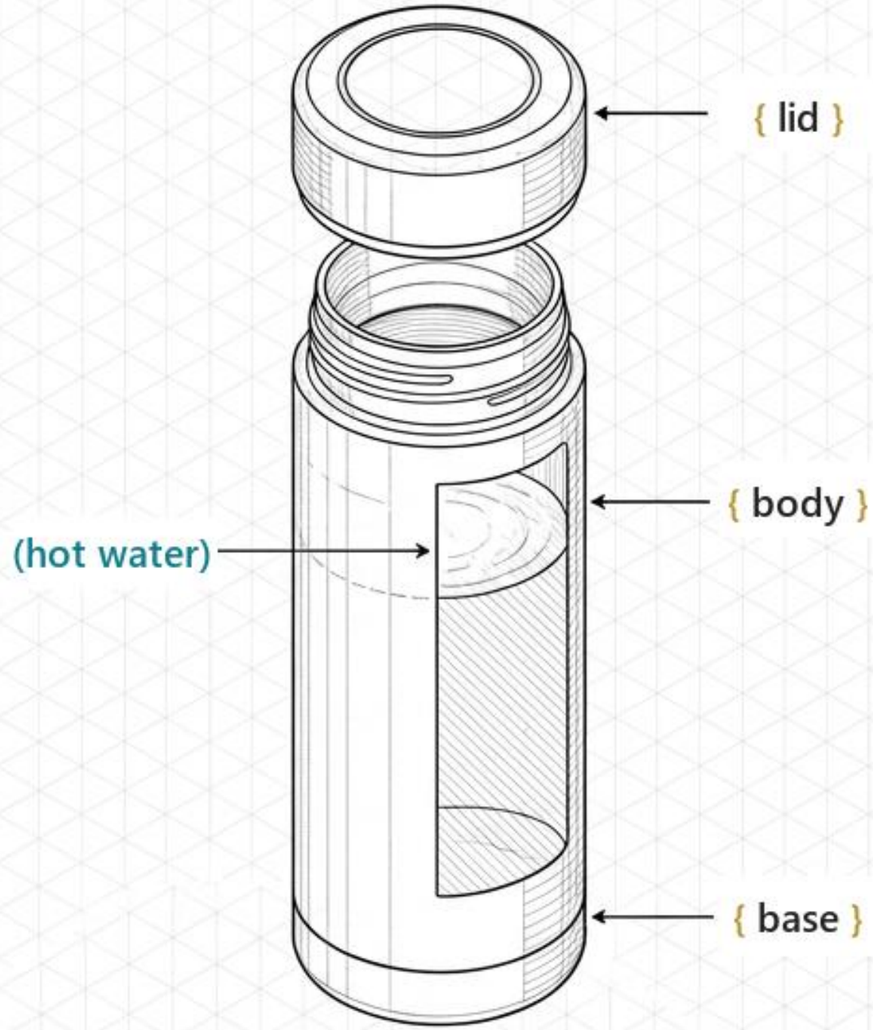
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The Proof Assembly Line: Putting the Three Supplements Back into the Six-Step Stairway



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Worked Example: The Complete Type-Membership Proof for a Thermos



[Theorem] thermos ∈ [cup]

Step 1–3 (Setup):

M = thermos. BOM = { lid, body, base }. S(M) = (hot water).

Step 4–5 (Reasoning via [ ]):

F(cup) = { can hold liquid, can be held and drunk from }.

f1 passes (holds liquid). f2 passes (can be held and drunk).

Step 6 (3-Layer Validation Check):

[Teal] Container layer: hot water → tea; id unchanged. (Check!)

[Coral] Composition layer: base essential; remove grading clear. (Check!)

[Gold] Abstract layer: ∩(T) fully passed. (Check!)

Conclusion:

All three layers pass → (Q.E.D.)

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Pitfall Guide: Five Common Misuses and the Precision-Fix Matrix

Misuse	Logical Failure	Fix
⚠ (water) written as { water }	Logical Failure: identity lost	✅ Fix: label id(C); put the contents in ( ).
⚠ [ ] written as a closed list	Logical Failure: new members cannot be classified	✅ Fix: use the open rule T = (F(T), ∩(T)).
⚠ Comparison instead of decomposition	Logical Failure: BOM confused with attributes	✅ Fix: strictly apply "detach and still recognize itself."
⚠ A new object whenever contents change	Logical Failure: container and content levels confused	✅ Fix: fix identity first, then discuss content replacement.
⚠ Concluding from a single layer	Logical Failure: logical blind spots	✅ Fix: run the joint three-layer check: container, composition, abstract.

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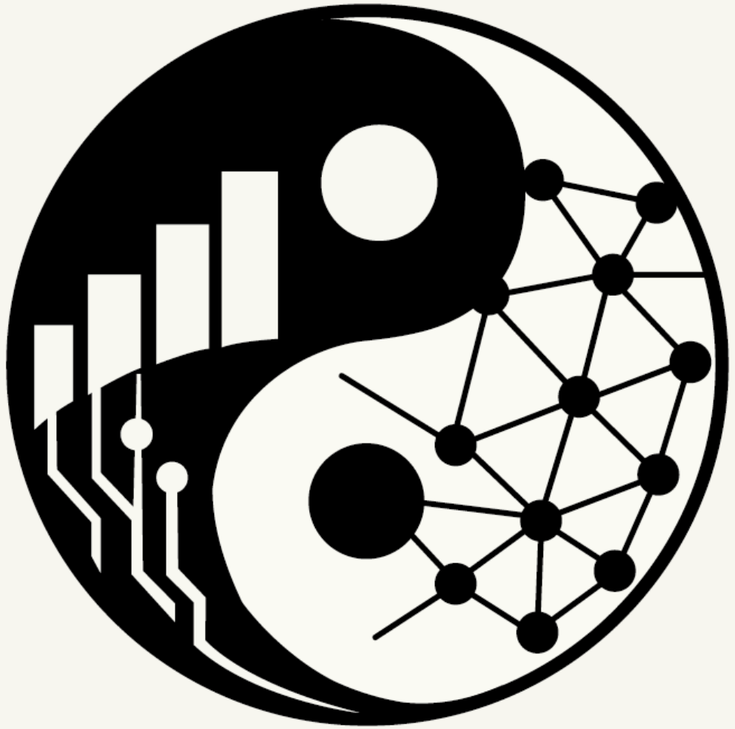
Final Acceptance: The 8-Point Checklist Before You Hand In a Proof

- [ ] 1. Object statement: Can one sentence name exactly which whole you are proving?
- [ ] 2. Definition sharpness: Is F(T) given up front, with each feature verifiable?
- [ ] 3. Bracket precision: Do { } parts, ( ) contents, [ ] category each keep their place?
- [ ] 4. Identity marker: For a container, is id(C) clearly marked?
- [ ] 5. Part grading: Have you sorted parts by remove (essential / downgraded / non-essential)?
- [ ] 6. Boundary test: Has at least one boundary case been judged correctly by the rule?
- [ ] 7. Step audit: Can a reader tick every sentence and reproduce your reasoning?
- [ ] 8. Three-layer closure: Container, composition, and abstract all pass before Q.E.D.?

A proof is precise because every layer lands exactly in the dimension where it belongs.

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LonGreat Vision: Creating Intelligent Life That Understands the Essence of the World



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"Let AI be more than a tool—a true intelligent life system capable of understanding, reasoning, learning, and creating."

From the unification of knowledge to the unification of thought, and ultimately to the unification of intelligence—this is the eternal path of AGI that LonGreat pursues.

www.longreat.net

Chang Changfeng
gamma@livemail.tw

Chang Tingying
mindaiworks@gmail.com